

MUHAMMAD USMAN

☎ (709) 730-3041 ✉ musman21@mun.ca [in linkedin.com/in/usman12529](https://www.linkedin.com/in/usman12529) github.com/musman12529 [Portfolio](#)

Education

Memorial University of Newfoundland

Bachelor of Science in Computer Science

Apr 2026

St. John's, NL

Technical Skills

Programming Languages: Python, JavaScript, Java, TypeScript

Frontend & Backend: React.js, Next.js, Node.js, Express.js, Spring Boot, REST APIs

Databases: PostgreSQL, MongoDB, MySQL, Neo4j

AI/LLM: RAG Pipelines, Vector databases (pgvector), OpenAI API, Embedding Models

Tools & Cloud/DevOps: Git, Postman, GitHub Actions, CI/CD, Azure, AWS (Certified Cloud Practitioner)

Work Experience

Founding Software Engineer

Jan 2026 – Present

Laniva AI

St. John's, NL

- Architected a SaaS platform using Next.js, PostgreSQL, and AWS automating property management operations across 350+ residential units
- Engineered backend services and REST APIs integrating OpenAI, telephony, SMS, and email to automate tenant communication and maintenance coordination, cutting manual effort by 60–70%
- Implemented a Retrieval-Augmented Generation (RAG) pipeline utilizing PostgreSQL with pgvector and structured Q&A segmentation, improving retrieval accuracy by 20%
- Deployed scalable AWS Elastic Beanstalk infrastructure with Application Load Balancing and GitHub Actions CI/CD pipelines, cutting manual deployment time from 15–20 minutes to a few minutes

Software Engineer

May 2025 – Apr 2026

Memorial University of Newfoundland, Department of History

St. John's, NL

- Designed and implemented Python ETL pipelines to migrate 200K+ records from MySQL to Neo4j, enabling efficient relationship queries while maintaining 100% data integrity
- Engineered a Node.js and Express.js backend integrated with Neo4j, exposing 30+ REST APIs for scalable graph querying
- Optimized Neo4j graph queries for complex relationships, reducing latency by 60% over legacy relational database queries
- Collaborated with a development team throughout the Agile Scrum lifecycle, contributing to sprint planning and delivery

Full Stack Developer

May 2025 – Dec 2025

Memorial University of Newfoundland, Department of Psychology

St. John's, NL

- Led development and deployment of a React-based web application serving 500+ students and faculty, improving accessibility and usability of departmental resources
- Implemented a Node.js backend with MongoDB, designing RESTful APIs to enable scalable and reliable data access
- Architected an admin management portal enabling faculty to manage site content and monitor activity independently, reducing ongoing developer maintenance
- Integrated Azure Blob Storage for media assets and managed deployment on Azure for production use

Projects

Premier League Fantasy | Java, Spring Boot, React.js, PostgreSQL, Python, Pandas

GitHub

- Engineered comprehensive data scraping of match statistics for 600+ players using Python and Pandas
- Built a machine learning model using scikit-learn to predict match winners from historical player and match data
- Developed an object-oriented Spring Boot backend with PostgreSQL and React.js to enable real-time data manipulation and dynamic presentation, streamlining user interaction and data flow

Project Management Platform | Next.js, TypeScript, Node.js, MongoDB, Socket.IO, Gemini API

GitHub

- Built a full-stack solution using Next.js, Node.js, and MongoDB, improving team coordination and task management
- Implemented real-time collaboration with Socket.IO and incorporated Gemini API for AI-driven task insights

Study Early | Flutter, Dart, Firebase, OpenAI API

GitHub

- Created a Flutter-based student productivity app for course management, scheduling, reminders, and progress tracking
- Integrated OpenAI for AI-generated quizzes and an in-app study assistant to improve learning